Denumerable sets

Def A set E is said to be *denumerable* if there exists a bijection $\mathbb{N} \longrightarrow E$.

Prop Every denumerable set is infinite.

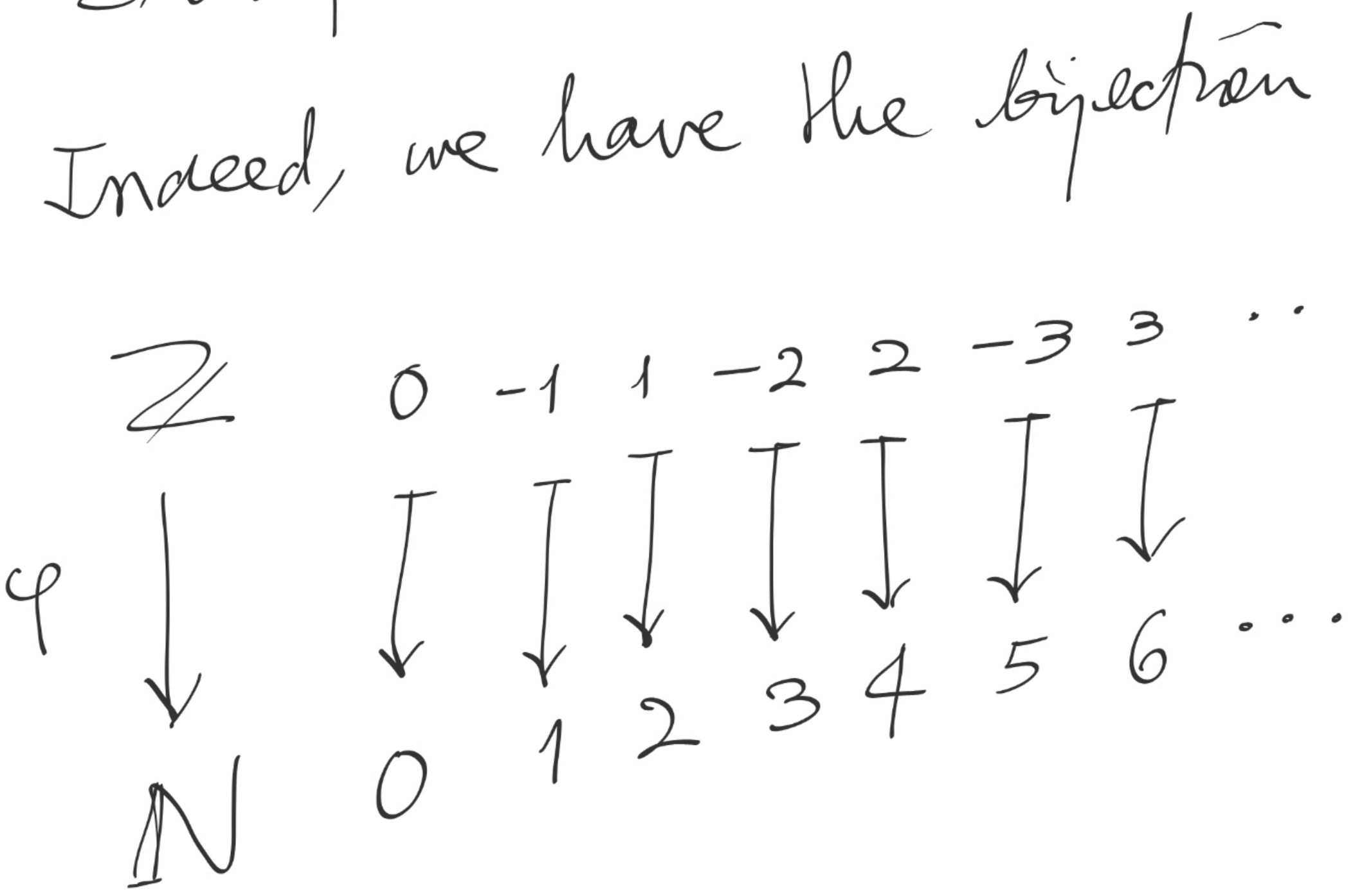
Proof. Let E be a denumerable set, $\varphi : \mathbb{N} \longrightarrow E$ a bijection, and $s : \mathbb{N} \longrightarrow \mathbb{N}$ the mapping $s(n) = n+1$. The composite

$$E \xrightarrow{\varphi^{-1}} \mathbb{N} \xrightarrow{s} \mathbb{N} \xrightarrow{\varphi} E$$

is injective but not surjective, hence E is infinite. $\square$

Example: The set $\mathbb{Z}$ is denumerable.

Indeed, we have the bijection



In other words

$$\varphi(n) = \begin{cases} 2n & \text{if } n \geq 0 \\ -2n-1 & \text{if } n < 0. \end{cases}$$

□

- Is every infinite set denumerable?

No. The set $\mathcal{P}(\mathbb{N})$ is denumerable because there is no surjection $\mathbb{N} \to \mathcal{P}(\mathbb{N})$, and it is infinite because there is an injection $\mathbb{N} \to \mathcal{P}(\mathbb{N})$ (for example $n \mapsto \{n\}$). □

We have used:

Let T be a set, $S \subset T$ a subset. If S is infinite, then T is infinite.

Proof. As S is infinite, there is a mapping $\alpha : S \to S$ which is injective but not surjective. Define $\beta : T \to T$ by

$$\beta(x) = \begin{cases} \alpha(x) & \text{if } x \in S \\ \end{cases}$$

$$\beta(x) = \begin{cases} \text{?} & \text{if } x \in S \\ x & \text{if } x \notin S. \end{cases}$$

Then β is injective but not surjective, hence T is infinite. □

Prop. 1 If E is denumerable and $\alpha : E \longrightarrow F$ is a bijection, then F is denumerable.

Proof Indeed, if $\varphi : \mathbb{N} \longrightarrow E$ is a bijection, then so is $\alpha \circ \varphi : \mathbb{N} \longrightarrow F$. □

Prop. 2 If E is denumerable and

$S \subset E$ is infinite, then

S is denumerable.

Proof. Let us first prove the special case $E = \mathbb{N}$. Define a mapping $\varphi : \mathbb{N} \to S$ as follows:

$\varphi(0) =$ the smallest element of S

$\varphi(1) =$ the smallest element of $S - \{\varphi(0)\}$

$\varphi(2) =$ the smallest element of $S - \{\varphi(0), \varphi(1)\}$

...

As φ is a bijection, S is denumerable.

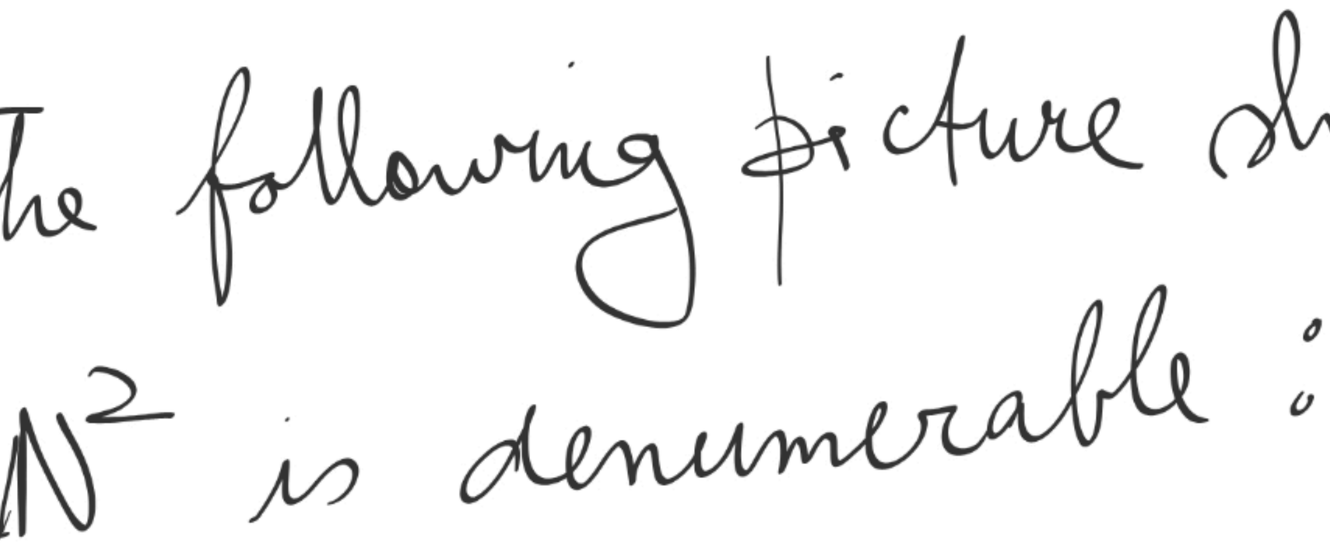
In general, let $\alpha : \mathbb{N} \to E$ be a bijection. Then $\alpha^*(S)$ is an infinite subset of $\mathbb{N}$, so it is denumerable. Also, α induces a bijection $\alpha^*(S) \to S$, so S is denumerable.

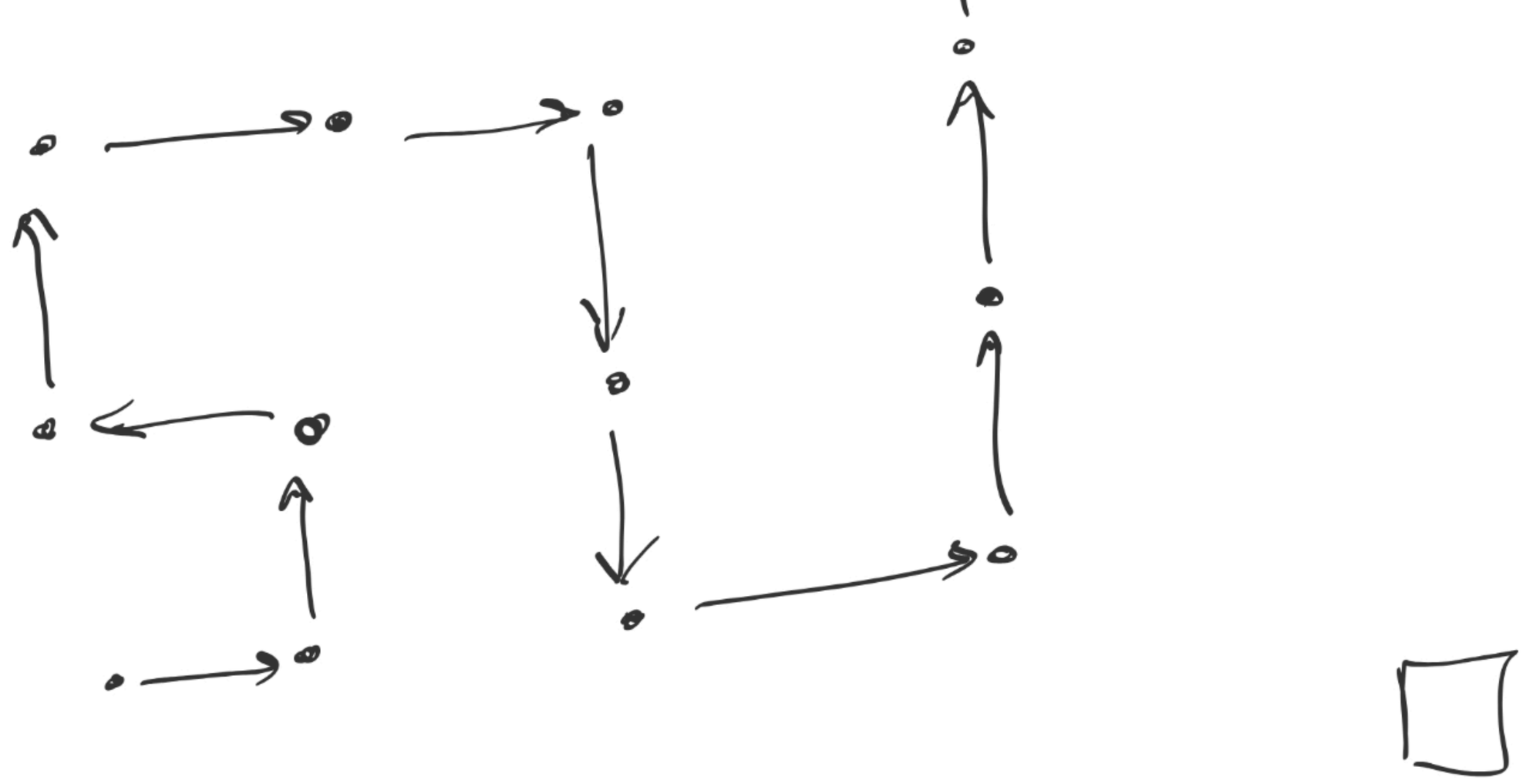
Prop 3 If S_1 and S_2 are denumerable, then so is $S_1 \times S_2$.

Proof. It is enough to prove that $\mathbb{N}^2$ is denumerable.

Indeed, if $\alpha_1 : \mathbb{N} \to S_1$ and $\alpha_2 : \mathbb{N} \to S_2$ are bijections, then so is $\alpha_1 \times \alpha_2 : \mathbb{N}^2 \to S_1 \times S_2$, defined by $(\alpha_1 \times \alpha_2)(x_1, x_2) = (\alpha_1(x_1), \alpha_2(x_2))$.

The following picture shows that $\mathbb{N}^2$ is denumerable:





Prop. 4 *If A is denumerable, $f : A \longrightarrow B$ is surjective, and B is infinite, then B is denumerable.*

Proof. As f is surjective, there exists a mapping $s : B \to A$ such that $f \circ s = Id_B$. Clearly s is injective:

$$s(x) = s(y) \Longrightarrow (f \circ s)(x) = (f \circ s)(y)$$

$$Id_B(x) = Id_B(y)$$

$\Rightarrow f_B(x) = f_B(y)$

$\Rightarrow x = y$.

It follows that $S_*(B)$ is an infinite subset of A, and s induces a bijection $S_*(B) \longrightarrow B$. By Prop. 2, $S_*(B)$ is denumerable, and hence B is denumerable. □

Theorem The set $\mathbb{Q}$ is denumerable.

Proof. 1. The set $\mathbb{Z}$ is denumerable. (see the Example above)

2. The set $\mathbb{Z}^2$ is denumerable. (Prop. 3).

3. The subset

$$E = \{(a,b) \in \mathbb{Z}^2 \mid b \neq 0\}$$

(Prop. 2)

is denumerable (by 3.)

4. $\mathbb{Q}$ is denumerable: indeed,

- $\mathbb{Q}$ is infinite (because $\mathbb{N} \subset \mathbb{Q}$ is infinite)
- $E \to \mathbb{Q}$, $(a,b) \mapsto a/b$ is surjective
- E is denumerable (by 3.)

Hence $\mathbb{Q}$ is denumerable, by Prop. 4.

□